

TOPIC 24/60

# What is Fine-Tuning?

Teaching a generalist  
to become a specialist.

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## THE ANALOGY

Pre-training teaches the AI the "world" (general knowledge). Fine-tuning teaches the AI a "trade" (specialized skill).

*"It's like taking a medical student who has read every textbook in the library (Pre-trained) and sending them through an intense residency program (Fine-tuned) to actually practice surgery."*

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## HOW IT WORKS



## Updating the Weights

We take an already "smart" base model and perform additional training on a smaller, highly curated dataset.

### The Goal:

Slightly adjust the model's internal neural weights so that it naturally optimizes for a specific style, task, formatting, or domain logic.

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WHEN TO USE IT

# Specialization

General foundation models are good at everything, but masters of nothing. Fine-tuned models are masters of one specific thing.

## Top Use Cases:

- Medical coding and diagnosis.
- Adopting a strict corporate brand voice.
- Legal document drafting & analysis.
- Understanding highly niche domain jargon.

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## THE PROCESS

# Curated Data

Quality over Quantity

Unlike pre-training (which ingests trillions of low-quality internet tokens), fine-tuning requires artisanal data.

You often only need **500 to 5,000** extremely high-quality, perfectly formatted prompt-and-response pairs to completely change a model's behavior.

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## THE TECHNIQUE

# Supervised

Pre-training is usually unsupervised (filling in the blanks). Fine-tuning is almost always supervised learning.

### Supervised Learning:

You show the AI exactly what the input is, and exactly what the perfect output looks like.

> **Input:** "Summarize this contract."

> **Output:** [Perfect legal summary]

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# THE DANGER ZONE

## THE PITFALL

### Catastrophic Forgetting

If you train the model too intensely on new, narrow data (like only writing Python code), it will physically overwrite its old weights.

> Result: It forgets how to speak English or answer general questions.

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## MODERN SOLUTIONS

# PEFT / LoRA

Training an entire 70-billion parameter model is incredibly expensive. Modern AI uses a trick called PEFT (Parameter-Efficient Fine-Tuning).

Instead of updating the main brain, we "freeze" the original model and only train tiny "adapter" layers (LoRA). This makes fine-tuning 1,000x cheaper and prevents Catastrophic Forgetting.

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# THE CHEAT SHEET

## Pre-training

Massive data (trillions of tokens), Unsupervised, extremely expensive, learns general world knowledge.

## Fine-tuning

Small data (thousands of examples), Supervised, relatively cheap, learns specific tasks and formatting.

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# Fine-Tuning

## CONCEPT UNLOCKED



## Goal: Specialized Performance



## Needs High-Quality Data



## Risk: Catastrophic Forgetting



## Made cheap using PEFT/LoRA

## CONNECT WITH ME



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